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Key Points

- Asia's cross-border payment architecture is becoming multilayered. Different systems, platforms and frameworks are likely to coexist rather than be replaced by a single technology.
- To expand digital trade or close the SME trade–finance gap, payment systems should be connected with digital identity systems among other services and facilities.
- Modernizing wholesale payment systems can make large-value cross-border transactions faster and more transparent. New arrangements are likely to complement rather than immediately replace correspondent banking.
- Policy makers should strengthen the institutional foundations of cross-border payments through the legal recognition of electronic trade documents and interoperable data among other standards and systems.

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Cross-border Digital Payments in Asia: Retail Connectivity, Wholesale Settlement, and Implications for Trade

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1. Introduction

Digital payments are becoming an increasingly important part of the infrastructure supporting trade, services, tourism, remittances, and financial integration in Asia. The most visible changes are occurring at the retail level, where cross-border linkages among fast payment systems and quick response (QR)-code networks allow consumers and businesses to make payments through familiar domestic banking or payment applications. These developments can reduce the cost, time, and complexity of small-value cross-border transactions.

Cross-border economic activity, however, involves much more than the transfer of money between a payer and a recipient. Even when an order, invoice, or service is delivered digitally, the transaction may require customer identification, foreign exchange conversion, regulatory compliance, trade documentation, logistics information, financing, and final settlement between financial institutions. Payment connectivity should therefore be examined as one component of a broader digital trade ecosystem.

This policy brief distinguishes between the retail and wholesale layers of cross-border payments. Retail payment connectivity provides the interfaces through which individuals, tourists, small merchants, and firms initiate and receive payments. Wholesale infrastructure supports correspondent banking, foreign exchange transactions, interbank obligations, large-value business-to-business (B2B) payments, and final settlement in central bank money. The two layers perform distinct but complementary functions. Retail payments are initiated through consumer-facing systems, while the resulting interbank obligations are cleared and ultimately settled through wholesale infrastructure.

Asia provides a particularly important setting for examining these developments. Retail payment connectivity has so far advanced more rapidly than wholesale payment reform. The Association of Southeast Asian Nations (ASEAN) member economies are expanding bilateral linkages among fast payment and QR-code systems while also developing multilateral arrangements, and India is extending the international connectivity of its Unified Payments Interface (UPI). The Republic

of Korea benefits from a relatively integrated domestic payment infrastructure, while Japan's payment market remains more fragmented despite progress in QR-code acceptance. The People's Republic of China (PRC) combines large private payment ecosystems centered on Alipay and WeChat Pay with the UnionPay network and emerging fast-payment connectivity.

At the wholesale level, correspondent banking and the Society for Worldwide Interbank Financial Telecommunication (SWIFT) messaging remain central to cross-border payments, while the International Organization for Standardization (ISO) 20022 is improving the standardization and data content of payment messages. Local currency transaction frameworks are also expanding in Asia, enabling participating banks to use bilateral currency pairs more directly while generally retaining correspondent banking and SWIFT messaging. More fundamental experiments, including Project Agorá and mBridge, are testing tokenized deposits and wholesale central bank digital currencies as alternative settlement arrangements. Private stablecoins provide another digital channel for cross-border transfers, although their predominantly United States (US) dollar denomination means that they do not currently represent a broad shift toward non-dollar settlement.

These innovations are especially relevant to small and medium-sized enterprises (SMEs). Retail connectivity can make it easier for an SME to receive small-value payments from foreign customers, while wholesale modernization can improve the banking and settlement processes supporting larger trade transactions. Nevertheless, payment speed alone cannot address the fixed costs of compliance, limited access to foreign exchange and hedging, inadequate collateral, fragmented trade documentation, or the shortage of working capital and trade finance. The benefits of payment innovation will depend on whether verified commercial, logistics, customs, and payment data can be connected and used to reduce transaction and financing frictions.

This policy brief argues that Asia is moving toward a multilayered cross-border payment architecture rather than a single regional system or form of digital money. Section 2 examines retail payment connectivity, while Section 3 analyzes the modernization of wholesale settlement. Section 4 considers how these developments interact with trade and SME finance, and Section 5 presents a sequenced policy agenda for Asia.

The central policy challenge is not simply to make payments faster. It is to ensure that multiple payment and settlement infrastructure can interact with one another and with the wider systems supporting trade. This requires legal recognition of electronic trade documents, common data standards, trusted digital identity, proportionate compliance, transparent foreign exchange and payment costs, competitive access, and effective cross-border governance. Without such coordination, technological innovation may create new liquidity silos and forms of fragmentation. With it, payment modernization can contribute to more inclusive digital trade and deeper regional economic integration.

2. Retail Cross-border Payments: Connectivity and Data Governance

Cross-border retail payment connectivity has become one of the most visible and advanced areas of Asia's payment transformation. This section examines retail payments as a layer of regional digital infrastructure supporting small-value cross-border economic activity. It first explains the functions of retail payment connectivity and then reviews the principal models emerging in Asia, including fast payment system linkages, QR-code interoperability, mobile payment ecosystems, and multilateral initiatives such as Project Nexus. It also examines the data-sharing, digital identity, compliance, and governance arrangements required to scale retail payment connectivity across multiple jurisdictions. Finally, it considers the potential roles of retail central bank digital currencies and privately issued stablecoins as alternative or complementary forms of digital money for cross-border payments.

2.1 Retail Payment Connectivity and the Digital Economy

The growing policy interest in retail payment connectivity should be understood in the context of the broader international agenda to improve cross-border payments. The Group of Twenty (G20) cross-border payments agenda seeks to make payments faster, less costly, more accessible, and more transparent (FSB 2021). These objectives are particularly important for retail payments and remittances, where high fees, delays, limited access, and uncertainty directly affect consumers, tourists, migrant workers, and SMEs, including small merchants.

In the digital economy, retail payments are no longer the final step in a retail or small-value transaction. They

are increasingly embedded in wider digital ecosystems involving e-commerce platforms, tourism services, logistics providers, digital identity systems, fraud monitoring, and compliance processes. A cross-border e-commerce transaction, for example, requires payment authorization and customer authentication, while the broader commercial transaction may also involve exchange rate disclosure, taxation, logistics, refunds, and dispute resolution. For this reason, the economic value of payment connectivity increases when it interacts reliably with other digital infrastructure.

Retail payment connectivity can support digital trade by reducing the cost and uncertainty of small-value cross-border transactions. Faster and cheaper retail payments can facilitate tourism spending, cross-border e-commerce, remittances, online services, and SME sales across borders. These functions are particularly relevant in Asia, given the importance of regional mobility, platform-based commerce, digital services, and small-value cross-border transactions.

At the same time, retail payment systems should not be seen simply as private wallet applications. In many Asian economies, mobile payment services are often linked to bank accounts and supported by fast payment systems, national QR standards, and common switching and clearing arrangements. Even when users initiate payments through mobile applications or QR codes, regulated financial institutions and payment system operators process the underlying transfers. Cross-border transactions also require foreign exchange conversion, settlement arrangements, and compliance controls. These institutional foundations determine whether retail payment services can be connected and scaled safely across borders.

Retail payment connectivity is therefore becoming part of the infrastructure supporting digital trade, regional mobility, and small-value cross-border economic activity. Its wider economic impact will depend on how effectively payment systems can connect with digital platforms, digital identity systems, data governance and compliance arrangements, and foreign exchange services.

2.2 *Models of Retail Payment Connectivity in Asia*

Retail payment connectivity in Asia is developing through domestic payment modernization, bilateral linkages, and emerging multilateral initiatives. The ASEAN member countries and India have made particularly visible progress, although through different models, including QR-based merchant payments, account-to-account fast payment linkages, and platform-based wallet networks. The key distinction lies not only in the applications used by consumers, but also in the underlying arrangements for transmitting payment instructions, transferring funds, converting currencies, and completing settlement.

ASEAN

ASEAN provides the clearest example of the regionalization of retail payment connectivity in Asia. The region has gradually moved from the development of domestic fast payment systems and national QR standards toward bilateral cross-border linkages. In 2022, the central banks of Indonesia, Malaysia, the Philippines, Singapore, and Thailand signed the Memorandum of Understanding on Cooperation in Regional Payment Connectivity, which subsequently expanded to include additional ASEAN central banks (Bank Indonesia 2022, 2024). Rather than establishing a single ASEAN-wide payment system, the Regional Payment Connectivity framework promotes interoperability among participating domestic payment infrastructure.

This approach was reinforced by the ASEAN Leaders' Declaration on Advancing Regional Payment Connectivity and Promoting Local Currency Transaction. The declaration calls for greater interoperability among cross-border payment systems, wider adoption of digital payments, stronger financial inclusion, and increased use of local currencies in regional transactions (ASEAN 2023). Payment integration in ASEAN is therefore developing through cooperation among central banks, payment system operators, banks, and regulated payment service providers rather than through the creation of a single regional payment platform.

Operationally, ASEAN retail payment connectivity is developing through two principal forms of cross-border connectivity: QR-based merchant payment linkages and account-to-account linkages between fast payment systems. These forms differ primarily in their user interfaces and functions, although both may rely on regulated banks, payment service providers, switching arrangements, foreign exchange conversion, and settlement infrastructure.

The first form consists of cross-border QR merchant payment linkages. QR codes provide the user-facing acceptance interface, allowing consumers to scan a foreign merchant's code using a familiar domestic banking or payment application. The underlying transaction is

processed through participating banks, payment service providers, switching arrangements, foreign exchange conversion, and settlement infrastructure. Cross-border QR payments should therefore not be understood simply as transactions between private mobile wallets; they are supported by regulated domestic payment systems and financial institutions.

Cross-border QR linkages have so far produced the most visible results in ASEAN. They allow tourists and other consumers to pay participating shops, restaurants, hotels, transport providers, and small merchants abroad using domestic payment applications. For example, the Indonesia–Singapore linkage connects the Quick Response Code Indonesian Standard (QRIS), Indonesia's national QR payment standard, with NETS QR, a merchant QR payment scheme in Singapore. Customers from Singapore can scan QRIS codes at participating merchants in Indonesia, while customers from Indonesia can scan NETS QR codes at participating merchants in Singapore (Bank Indonesia and MAS 2023). Such arrangements can reduce reliance on cash, simplify foreign currency payments, and broaden access to cross-border payment acceptance, particularly for micro, small, and medium-sized merchants.

The second form of connectivity consists of cross-border linkages between fast payment systems, which primarily support direct account-to-account transfers. Under this model, funds can be sent directly to an individual's or business account using an account number, mobile phone number, payment address, or another proxy identifier. Unlike merchant QR payments, which are initiated by scanning a merchant's code, account-to-account transfers can be made without a merchant payment interface and are particularly relevant for person-to-person transfers and remittances. They require arrangements for beneficiary identification, transaction limits, erroneous payments, anti-money-laundering and countering-the-financing-of-terrorism controls, foreign exchange disclosure, and cross-border settlement.

Although merchant QR payment linkages are more widespread across ASEAN, account-to-account fast payment linkages are already operational in some bilateral corridors. The PayNow–PromptPay linkage between Singapore and Thailand, launched in 2021, enables individuals to make real-time cross-border transfers using registered mobile phone numbers (MAS and BOT 2021). The PayNow–DuitNow linkage between Singapore and Malaysia, launched in 2023, enables customers of participating financial institutions to make real-time cross-border transfers (MAS and BNM 2023). These arrangements demonstrate that linkages between fast payment systems can support person-to-person transfers and remittances as well as payments to businesses.

ASEAN's cross-border retail payment connectivity has therefore advanced through a combination of merchant QR corridors and selected account-based fast payment linkages. However, their geographical coverage and functionality continue to differ. Merchant QR payments are available across a growing number of bilateral corridors, whereas real-time account-to-account transfers remain concentrated in a smaller number of country pairs. ASEAN's experience illustrates how bilateral connections among domestic payment infrastructure are gradually developing into a broader regional payment network, even though users cannot yet access the same range of services across all participating economies.

Figure 1 summarizes the main bilateral retail payment linkages within ASEAN and distinguishes QR-based merchant payment connectivity from account-to-account linkages between fast payment systems.

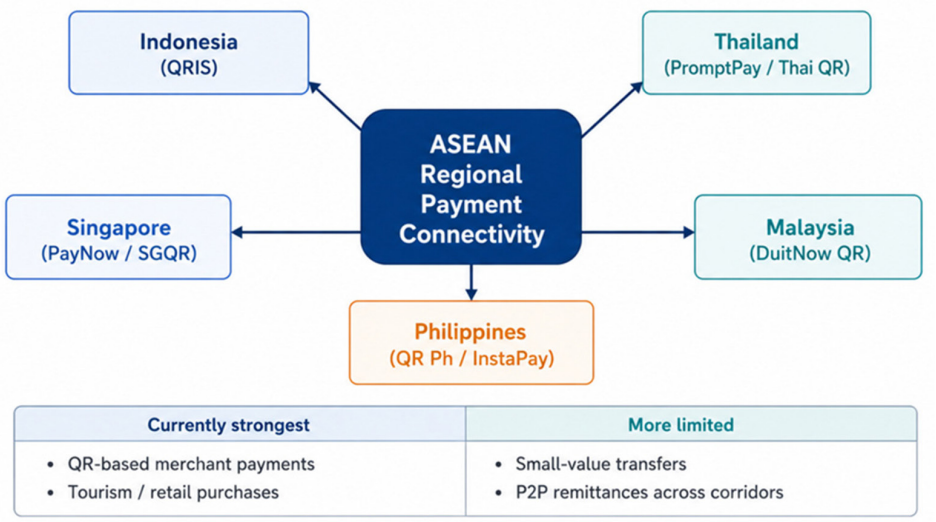
India

India provides a useful illustration on how a domestic account-based fast payment system can support both merchant payments and direct transfers between individuals. The UPI is an instant payment system developed and operated by the National Payments Corporation of India, an entity regulated by the Reserve Bank of India (Digital India n.d.). The UPI enables immediate interbank transfers through mobile applications linked to bank accounts and supports both person-to-person (P2P) transfers and person-to-merchant payments.

Users can initiate UPI transactions through QR codes or identifiers such as mobile phone numbers, UPI identification, and virtual payment addresses, without entering the recipient's full bank account details. Although the transaction may be initiated through a bank or third-party mobile application, the funds are generally transferred between accounts held at regulated financial institutions. The UPI therefore combines a convenient mobile interface with account-based payment infrastructure.

The international expansion of the UPI has followed two principal paths. The first is the cross-border linkage of account-based payment services for small-value

Figure 1: Intra-ASEAN Cross-border Payment Connectivity



ASEAN = Association of Southeast Asian Nations, P2P = person-to-person, QR = quick response, QRIS = Quick Response Code Indonesian Standard.

Source: Author's illustration based on Bank Indonesia (2022, 2024), MAS and BOT (2021), Bank Indonesia and MAS (2023), and MAS and BNM (2023).

transfers and remittances. The UPI–PayNow linkage, launched in 2023, connects India's UPI with Singapore's PayNow instant transfer service and enables customers of participating financial institutions to send funds in both directions. The service is designed to provide faster and less costly remittances than conventional cross-border transfer channels. However, it is not a general-purpose cross-border transfer service: eligible transfers are currently limited to P2P remittances to relatives abroad and for gifts (RBI 2023).

The second path is the expansion of UPI-based merchant payment acceptance outside India. UPI Global Acceptance enables eligible Indian users to make QR-based payments at participating merchants in selected overseas markets using supported UPI applications (NIPL n.d.). This service is mainly relevant to Indian travelers making purchases abroad. It should be distinguished from the UPI–PayNow linkage because it enables payments to foreign merchants rather than direct transfers to an individual's account.

India is therefore extending the same domestic payment infrastructure internationally through arrangements that perform different functions. India's experience demonstrates that an account-based fast payment system can support both cross-border transfers and merchant payments, although their regulatory requirements, participating institutions, geographical coverage, and eligible uses differ.

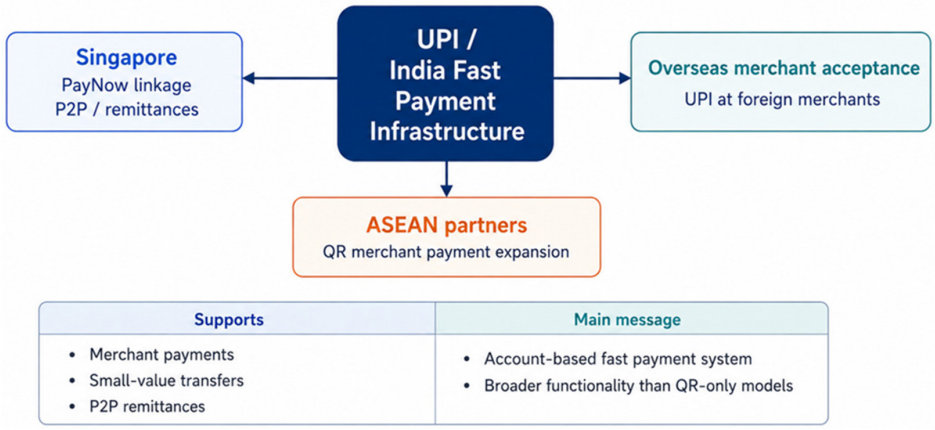
Figure 2 summarizes these two paths of the UPI's international expansion: account-based cross-border remittances through the UPI–PayNow linkage and QR-based payment acceptance at participating overseas merchants.

The Republic of Korea

The Republic of Korea provides a useful comparative case because it combines multiple private payment providers with relatively centralized interbank retail payment infrastructure. Its retail payment market includes banks, card companies, mobile payment services, and fintech providers. However, most major interbank retail payment systems are operated by the Korea Financial Telecommunications and Clearings Institute (KFTC). The KFTC conducts clearing through systems such as the Giro System and Interbank Shared Networks, while the resulting net obligations among participating financial institutions are ultimately settled in central bank money through accounts at the Bank of Korea using BOK-Wire+ (Bank of Korea n.d.).

This institutional structure does not mean that all mobile wallets, merchant contracts, fees, and customer networks in the Republic of Korea are unified. It nevertheless provides a common interbank infrastructure through which banks and payment service providers can exchange payment instructions and clear retail transactions. This distinguishes the Republic of Korea from markets in

Figure 2: India's Cross-border Retail Payment Connectivity



ASEAN = Association of Southeast Asian Nations, P2P = person-to-person, QR = quick response, UPI = Unified Payments Interface.

Source: Author's illustration based on RBI (2023), Digital India (n.d.), and NIPL (n.d.).

which the underlying payment infrastructure, as well as the customer-facing services, is more fragmented across private providers.

The Republic of Korea's cross-border QR payment linkage with Indonesia illustrates how domestic payment infrastructure can be connected to an overseas payment ecosystem. Following preparations announced by the Bank of Korea in February 2026, the Republic of Korea and Indonesia launched the linkage on 1 April 2026 (Bank of Korea 2026; Bank Indonesia 2026). Customers of participating institutions can use supported domestic applications to make QR payments at participating merchants in the other economy. The arrangement involves Bank Indonesia, the Bank of Korea, the Indonesia Payment System Association (ASPI), the KFTC, payment infrastructure providers, and participating financial institutions. It therefore connects domestic payment infrastructure and participating institutions rather than simply linking one private wallet with another.

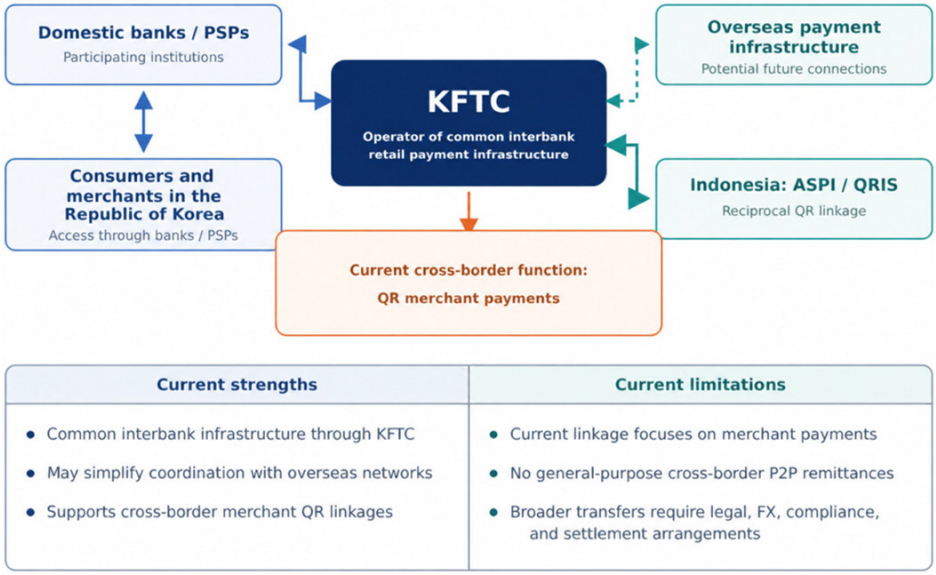
The operational linkage focuses on QR-based merchant payments. It allows consumers and travelers to make purchases at participating foreign merchants through familiar domestic applications, but it does not provide a general cross-border person-to-person remittance service comparable to UPI–PayNow. Establishing such a service would require additional arrangements for beneficiary identification, account or proxy identifiers, transaction limits, erroneous payments, compliance controls, foreign exchange conversion, and cross-border settlement.

The KFTC describes K-Payment as a centralized payment and settlement system model and includes cross-border fund transfer services among its international applications (KFTC n.d.). This suggests that the Republic of Korea's relatively centralized interbank infrastructure could facilitate the future expansion of cross-border retail payment services. However, domestic technical capacity alone is not sufficient. Broader account-to-account transfers would also require agreements concerning eligible transactions, participating institutions, foreign exchange arrangements, anti-money laundering and countering-the-financing-of-terrorism (AML/CFT) controls, consumer protection, and settlement.

The Republic of Korea's current model therefore combines common domestic interbank infrastructure with selective cross-border QR merchant payment connectivity. Its institutional structure may make coordination with overseas payment systems easier than in a highly fragmented domestic market, but cross-border functionality remains more limited than domestic functionality. In particular, the Indonesia–Republic of Korea linkage should be distinguished from bilateral fast payment linkages that already support direct cross-border P2P remittances.

Figure 3 illustrates the role of the KFTC as a common domestic infrastructure operator and the current emphasis of the Republic of Korea's cross-border connectivity on QR-based merchant payments.

Figure 3: Republic of Korea's Cross-border Retail Payment Connectivity



ASPI = Indonesia Payment System Association, FX = foreign exchange, KFTC = Korea Financial Telecommunications and Clearings Institute, P2P = person-to-person, PSP = payment service provider, QR = quick response, QRIS = Quick Response Code Indonesian Standard.

Source: Author's illustration based on Bank of Korea (n.d. 2026), Bank Indonesia (2026), and KFTC (n.d.).

Japan

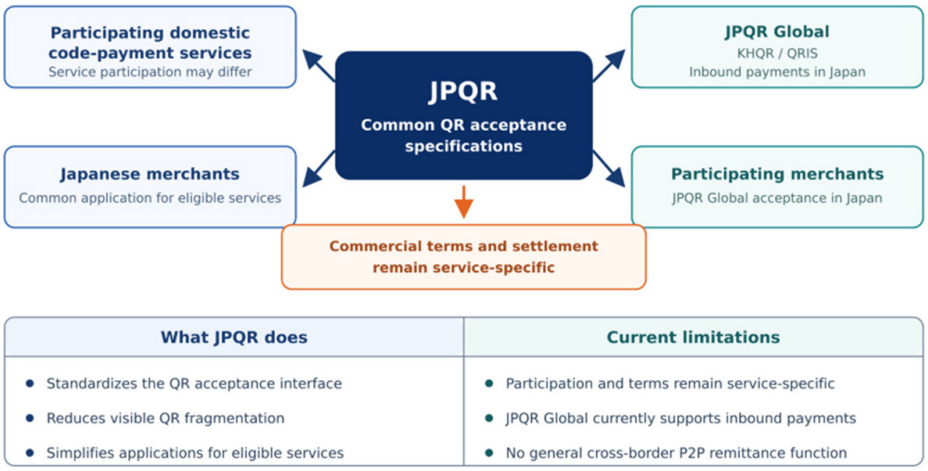
Japan provides a contrasting case because its retail payment market has become increasingly digital, while its domestic payment infrastructure remains divided among multiple code-payment services, bank transfer systems, card networks, and other payment providers. Major code-payment services include PayPay, NTT DOCOMO's d Payment, Rakuten Pay, and au PAY. Japan also has well-developed domestic bank transfer infrastructure, but merchant code payments and account-to-account transfers operate largely through separate arrangements. Japan therefore does not yet have a common retail payment infrastructure that supports both cross-border merchant payments and cross-border small-value transfers.

JPQR was introduced in 2019 as a common specification for payment QR codes and barcodes. Developed by the Payments Japan Association, it allows participating payment services to use a standardized acceptance interface. Merchants can apply for multiple eligible services through a common application process and can use a common interface to review sales information for participating services. JPQR can therefore reduce the visible proliferation of separate QR codes at merchants and lower some of the administrative burden associated with accepting multiple code-payment services (Payments Japan Association 2025a).

JPQR should not, however, be understood as a unified payment system or settlement infrastructure. The underlying payment services remain provider-specific, and merchants must select the services they wish to accept. PayPay, one of Japan's largest code-payment services, is not currently listed among the payment services participating in the JPQR common merchant application framework. Applicable fees, settlement cycles, contractual terms, customer networks, and other operating conditions continue to differ across providers. A merchant displaying a JPQR code therefore does not automatically accept every domestic code-payment service. JPQR standardizes part of the merchant acceptance interface but does not integrate the underlying payment providers, customer accounts, clearing arrangements, or settlement processes.

Japan began extending this acceptance framework across borders through JPQR Global in 2025. Cambodia's KHQR became available through JPQR Global in July 2025, followed by Indonesia's QRIS in August 2025 (Payments Japan Association 2025b, 2025c). These arrangements enable eligible Cambodian and Indonesian users to scan JPQR Global codes at participating merchants in Japan using supported home-country applications. In October 2025, the Payments Japan Association began accepting applications from Japanese merchants wishing to introduce JPQR Global, extending the service

Figure 4: Japan's Cross-border Retail Connectivity



KHQR = Cambodia's quick response code payment system, P2P = person-to-person, QR = quick response, QRIS = Quick Response Code Indonesian Standard.
Source: Author's illustration based on Payments Japan Association (2025a, 2025b, 2025c, 2025d).

beyond its initial implementation at selected locations (Payments Japan Association 2025d).

The current JPQR Global arrangements primarily support inbound merchant payments in Japan rather than general-purpose cross-border transfers. Foreign visitors using eligible applications can pay participating Japanese merchants, but JPQR Global does not enable general P2P remittances or account-to-account transfers. A common QR specification can improve merchant acceptance, but it does not by itself provide the beneficiary identification, account or proxy resolution, compliance controls, error handling, and settlement arrangements required for cross-border remittances. Japan's model therefore represents an expansion of cross-border QR acceptance rather than an integration of merchant payments and account-to-account transfers. Its wider contribution to regional connectivity will depend on merchant coverage, participation by domestic providers, and stronger interoperability with account-based payment infrastructure.

Compared with the Republic of Korea, Japan has a less centralized common infrastructure layer for interbank retail payments. The KFTC operates much of the Republic of Korea's common interbank retail payment infrastructure, whereas JPQR primarily standardizes the acceptance interface among participating code-payment services. Japan would therefore need to strengthen interoperability not only with overseas payment systems but also among its domestic merchant-payment and account-based transfer arrangements.

Figure 4 summarizes the role of JPQR as a common QR acceptance specification, the extension of this framework through JPQR Global, and the continuing separation between merchant acceptance and cross-border remittance functions.

The People's Republic of China

The PRC represents a different model of retail payment connectivity from the public infrastructure and fast-payment linkages examined in the preceding cases. Its domestic mobile payment market is organized largely around major private platforms, particularly Alipay and Weixin Pay, commonly known outside the PRC as WeChat Pay. These applications combine payment functions with e-commerce, messaging, transport, merchant services, and other digital activities. Payment connectivity has therefore developed through large platform ecosystems as well as through conventional banking and card infrastructure.

The international expansion of Alipay and Weixin Pay initially focused on enabling users from the PRC to pay overseas merchants through familiar domestic applications. Weixin Pay, for example, has connected its merchant-payment services with overseas acceptance infrastructure, including Malaysia's PayNet, Thailand's PromptPay, and Singapore's SGQR (Tencent 2026). Such arrangements expand overseas merchant acceptance but do not by themselves create a general account-to-account transfer system between users in the PRC and other Asian economies.

Alipay+ represents a related but distinct model. Alipay is a digital wallet used primarily in the PRC, whereas Alipay+ is an international wallet gateway operated by Ant International. It allows participating merchants and acquirers to accept multiple wallets and mobile payment applications through a common integration (Alipay+ n.d.). Participating applications include Alipay and AlipayHK, as well as Touch 'n Go eWallet from Malaysia, GCash from the Philippines, Kakao Pay from the Republic of Korea, and TrueMoney from Thailand. In Japan, users of participating applications can make payments at eligible PayPay merchants without adopting a separate Japanese payment application (Ant International 2024). Alipay+ therefore facilitates the cross-border use of domestic wallets, but it is a privately operated merchant-payment gateway rather than an interlinking of domestic fast payment systems comparable to Project Nexus.

UnionPay performs another function. It is an international payment network supporting cards, online payments, mobile applications, ATM access, and QR payments. UnionPay International works with overseas issuers, financial institutions, wallets, acquirers, and national payment networks. Its QR interoperability arrangements include connections that allow UnionPay users to pay at DuitNow QR acceptance points in Malaysia and PromptPay acceptance points in Thailand (UnionPay International 2023, 2025). UnionPay therefore combines the functions of an international card network with an expanding cross-border QR acceptance network.

These arrangements increasingly support two-way merchant acceptance. Users from the PRC can use Alipay, Weixin Pay, or UnionPay at participating overseas merchants, while selected overseas wallets and internationally issued cards can be used at participating merchants in the PRC. The model nevertheless remains centered primarily on merchant payments associated with tourism, retail commerce, and e-commerce. It differs from fast-payment linkages such as UPI–PayNow, which directly support eligible account-to-account transfers between participating users.

Cross-border remittance services linked to the PRC's payment networks also exist. TenPay Global provides selected remittance services connected with the Weixin ecosystem, while UnionPay's cross-border remittance services operate through participating institutions and corridors (TenPay Global n.d.; UnionPay International n.d.). These services, however, remain provider- and corridor-specific and do not constitute a common multilateral fast-payment network linking the PRC with other Asian economies.

Direct fast-payment-system connectivity is also emerging. Payment Connect, launched in June 2025, links the PRC's internet banking payment system with Hong Kong, China's Faster Payment System. It enables participating residents and institutions to make real-time cross-boundary remittances through participating banks, subject to applicable eligibility, foreign exchange, and transaction requirements (PBOC and HKMA 2025). Payment Connect is important because it extends the PRC's cross-border retail payment connectivity beyond merchant wallets and payment networks to account-based transfers. Its geographical scope, however, remains more limited than the multilateral architecture envisaged under Project Nexus.

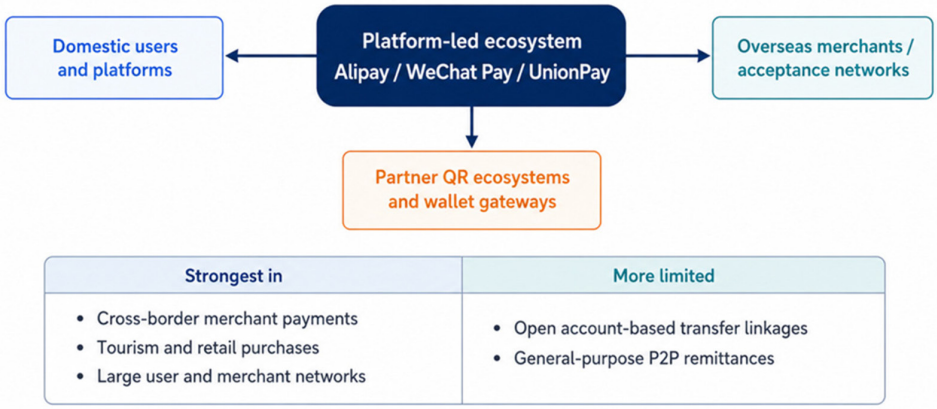
The PRC model is therefore best understood as a hybrid platform-and-infrastructure model. Large private wallet ecosystems provide scale, convenience, and extensive merchant reach; Alipay+ connects multiple domestic wallets with overseas merchants; UnionPay provides an international card and QR interoperability network; and Payment Connect introduces direct fast-payment-system connectivity between the PRC and Hong Kong, China. These arrangements demonstrate that regional payment connectivity can develop through several institutional channels, although cross-border account-to-account transfers involving the PRC remain more geographically and institutionally limited than cross-border merchant payments.

Figure 5 summarizes the PRC's hybrid model of cross-border retail payment connectivity, in which private wallet ecosystems, international payment networks, cross-border remittance services, and emerging fast-payment-system linkages perform different but complementary functions.

2.3 *Multilateral Retail Payment Connectivity: Project Nexus and Supporting Governance*

As cross-border retail payment linkages expand, the policy challenge is shifting from the establishment of individual bilateral corridors to the development of scalable multilateral arrangements. Such arrangements require two complementary elements. The first is a common architecture that allows domestic payment systems to connect without constructing a separate link for every participating jurisdiction. The second is an institutional framework that enables payment data, customer identification, compliance controls, and

Figure 5: The People's Republic of China's Cross-border Payment Connectivity



P2P = person-to=person, QR = quick response.

Source: Author's illustration based on Alipay+ (n.d.), Ant International (2024), Tencent (2026), UnionPay International (2023, 2025, n.d.), TenPay Global (n.d.), and PBOC and HKMA (2025).

operational responsibilities to function consistently across borders. Project Nexus illustrates the development of the common interlinking architecture, while data governance, identity, and compliance constitute the supporting foundations needed to make multilateral connectivity secure and reliable.

From Bilateral Linkages to Project Nexus

The development of retail payment connectivity in ASEAN and India illustrates both the benefits and the limitations of bilateral payment linkages. Bilateral arrangements can provide a practical starting point because two jurisdictions can coordinate technical standards, foreign exchange arrangements, settlement procedures, compliance requirements, and operational rules. However, this model becomes increasingly difficult to scale as the number of participating economies grows. Each additional corridor requires separate technical connections, contractual arrangements, operating procedures, liability rules, and supervisory coordination. A network based exclusively on bilateral linkages may therefore become fragmented and costly to expand.

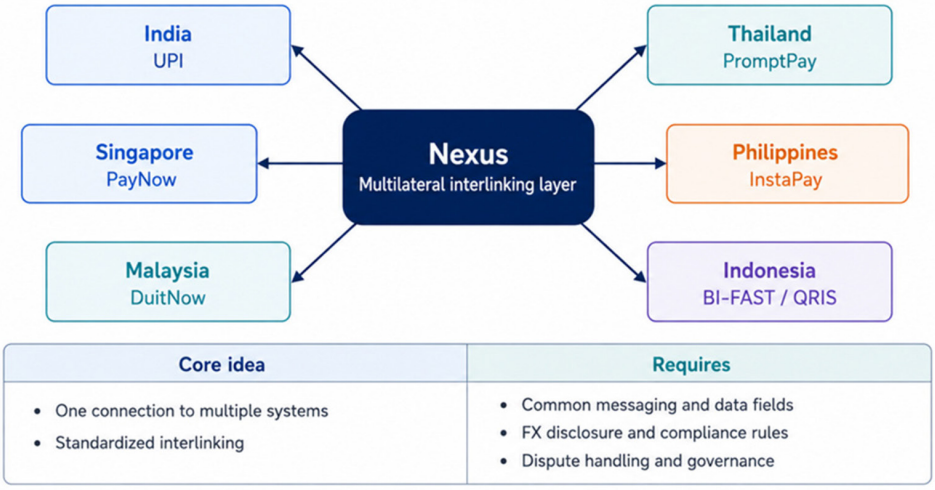
Project Nexus seeks to address this scalability problem through a multilateral framework for interlinking domestic instant payment systems. Rather than requiring each payment system operator to build a separate connection with every other participating system, the Nexus model envisages that an operator would establish a single connection to the Nexus platform and thereby gain access to the other systems connected to the network. Its significance therefore lies not only in faster payment execution but also in the possibility of establishing common technical standards, operating rules, and governance arrangements for cross-border retail payments (BIS Innovation Hub 2024b).

Nexus does not replace domestic instant payment systems. Each participating economy retains its own domestic payment infrastructure, regulatory framework, and participating financial institutions. Nexus instead standardizes how participating systems communicate and process cross-border payments. Its framework covers governance, scheme rules, technical architecture, ISO 20022 messages, foreign exchange arrangements, and the respective roles of payment system operators and participating financial institutions. In this sense, Nexus represents a shift from individually negotiated bilateral corridors toward a network-based model.

Project Nexus has progressed beyond its initial conceptual and experimental stages. The BIS Innovation Hub completed a comprehensive blueprint in 2024 covering the scheme's governance, commercial model, technical architecture, and operating arrangements. In 2025, the participating central banks established Nexus Global Payments as a not-for-profit entity in Singapore to operationalize and manage the scheme. The initiative is therefore moving toward live implementation, although it should not yet be described as a fully operational regional payment network (BIS 2026b).

The participating economies include India, Indonesia, Malaysia, the Philippines, Singapore, and Thailand. Their relevant domestic instant payment systems include

Figure 6: Project Nexus: One Connection to Multiple Instant Payment Systems



FX = foreign exchange, QRIS = Quick Response Code Indonesian Standard, UPI = Unified Payments Interface.

Note: Project Nexus aims to move from separate bilateral corridors toward a common multilateral framework for cross-border retail payments.

Source: Author's illustration based on BIS Innovation Hub (2024b) and BIS (2026b).

India's Unified Payments Interface, Indonesia's BI-FAST, Malaysia's Real-time Retail Payments Platform, which supports DuitNow services, the Philippines' InstaPay, Singapore's FAST, which supports PayNow, and Thailand's PromptPay infrastructure.

Figure 6 illustrates the central principle of the Nexus model in which each domestic instant payment system connects to a common platform to reach other participating systems.

Data Governance, Identity, and Compliance

Multilateral retail payment connectivity requires more than technical links among domestic payment systems. A cross-border payment also requires the transmission and verification of information about the payer, payee, account or wallet identifier, transaction amount, currency conversion, fees, and payment purpose. Participating institutions must be able to process this information consistently and use it to meet operational and regulatory requirements. Payment interoperability therefore depends on data interoperability as well as the movement of funds.

These requirements become more complex when connectivity extends beyond merchant QR-code payments to account-to-account transfers and remittances. In a merchant payment, the recipient is generally a registered merchant, and many relevant checks can be incorporated into merchant onboarding, acquiring arrangements, and the rules of the participating payment service providers. Account-to-account transfers require additional arrangements for identifying the beneficiary, validating an account or proxy identifier, applying transaction limits, preventing fraud, handling erroneous payments, and meeting AML/CFT requirements. Participating institutions must also have sufficient information to conduct applicable sanctions screening and transaction monitoring.

Common messaging and data standards are therefore essential for multilateral payment connectivity. If institutions collect, transmit, or interpret payment information differently, cross-border transactions may require manual intervention and become slower, more costly, and more difficult to monitor. The Committee on Payments and Market Infrastructures harmonized ISO 20022 data requirements to establish a common minimum set of information for cross-border payments, improve straight-through processing, and support compliance (CPMI 2023). Project Nexus similarly incorporates ISO 20022 message and application programming interface specifications into its technical framework (BIS Innovation Hub 2024b). This illustrates that a common connection among payment systems must be supported by common rules for structuring and exchanging payment data.

Customer identification and authentication are closely related to this issue. Domestic payment systems generally rely on customer due diligence conducted when an account or wallet is opened, together with authentication procedures, transaction limits, and ongoing monitoring. In a cross-border network, these domestic controls must operate across jurisdictions with different legal frameworks and supervisory practices. This does not necessarily require the creation of a single regional digital identity system. It does, however, require participating institutions and authorities to have confidence that customers have been appropriately identified, transactions are properly authorized, and relevant information can be made available when legally required.

At the same time, multilateral payment connectivity should not become an unrestricted data-sharing arrangement. Participating systems should exchange only the information necessary to process payments, manage risks, meet legal obligations, and resolve disputes. Appropriate safeguards include purpose limitation, data minimization, cybersecurity, auditability, restrictions on secondary use, and clear rules governing regulatory access. These safeguards are particularly important in Asia, where jurisdictions differ in their privacy rules, data-localization requirements, and approaches to cross-border transfers of personal and commercial information.

A clear allocation of responsibilities is equally important. Multilateral arrangements must specify which institutions are responsible for customer due diligence, sanctions screening, fraud management, foreign exchange disclosure, error handling, dispute resolution, and compensation when a payment is misdirected or fails. They must also establish procedures for operational incidents and cyberattacks. As the number of participating jurisdictions increases, common governance and liability rules become as important as the technical standards connecting the payment systems.

These institutional foundations also link payment connectivity to digital trade. Cross-border commerce may require payment information to interact with electronic invoices, e-commerce platforms, tax and customs procedures, logistics data, and digital identity services. For SMEs, including small merchants, reliable links among these systems can reduce administrative costs and uncertainty. The wider economic value of Project Nexus and similar initiatives will therefore depend not only on whether payments can move quickly across borders, but also on whether data, identity, compliance, and governance arrangements can support a secure and trusted multilateral network.

2.4 *Implications for Digital Trade and Regional Integration*

Asia's cross-border retail payment architecture is developing through multiple models rather than a single regional template. ASEAN and India are expanding connectivity through QR merchant payments and account-based fast payment linkages; the Republic of Korea builds on relatively integrated domestic interbank infrastructure; Japan has extended cross-border QR acceptance despite domestic fragmentation; and the PRC combines large platform ecosystems with UnionPay and emerging fast-payment connectivity. These differences reflect the structure of each economy's domestic payment infrastructure and regulatory framework.

The distinction between merchant payments and account-to-account transfers remains important. QR and wallet arrangements are particularly useful for tourism and retail purchases, whereas fast-payment linkages can also support P2P transfers and remittances but require additional arrangements for beneficiary identification, compliance, foreign exchange conversion, error handling, and settlement. Project Nexus seeks to scale such linkages through common technical standards and operating rules, but its success will also depend on data governance, digital identity, liability arrangements, regulatory cooperation, and cyber resilience.

Retail connectivity can support tourism, remittances, e-commerce, digital services, and SME sales, particularly when payment systems interact with electronic invoices, trade platforms, identity systems, foreign exchange services, and compliance processes. Retail central bank digital currencies (CBDCs) and stablecoins could provide additional options, but they do not currently drive regional integration: cross-border retail CBDCs remain experimental, while stablecoins are predominantly denominated in US dollars (BIS 2026a). In the near term, greater interoperability among existing regulated payment systems is likely to remain the principal source of progress.

3. Wholesale Settlement Infrastructure and Digital Innovation

While the retail payment innovations examined in Section 2 have transformed the user-facing side of cross-border payments, the underlying transfer of funds continues to depend substantially on banks, foreign exchange providers, correspondent accounts, and wholesale settlement systems. The infrastructure supports interbank transfers, the cash leg of securities and foreign exchange transactions, large-value corporate and trade-related payments, and final settlement in central bank money. Given their systemic importance, they are generally operated or overseen by central banks and other financial regulators to safeguard financial stability, support liquidity management, and ensure settlement finality. At the same time, digitalization, demand for faster cross-border settlement, operational inefficiencies, geopolitical fragmentation, sanctions risk, and concerns about financial resilience are creating pressure for change. This section first examines correspondent banking, SWIFT messaging, and local currency transaction frameworks; it then considers the modernization of existing infrastructure through SWIFT Global Payments Innovation (GPI), a service that improves the speed, transparency, and end-to-end tracking of cross-border payments, and ISO 20022, followed by tokenized deposits, Project Agorá, and wholesale central bank digital currency arrangements. It concludes by considering how existing and emerging mechanisms may coexist within a multilayered wholesale settlement architecture.

3.1 Correspondent Banking, SWIFT Messaging, and Local Currency Transactions

The traditional architecture of wholesale cross-border payments is based on correspondent banking relationships and the SWIFT financial messaging network. Despite the emergence of new technologies and alternative settlement arrangements, this architecture continues to form the backbone of cross-border bank payments. It enables banks to make payments in currencies and jurisdictions in which they do not have direct access to the relevant domestic payment or settlement system.

Conventional Cross-border Payment through Correspondent Banking

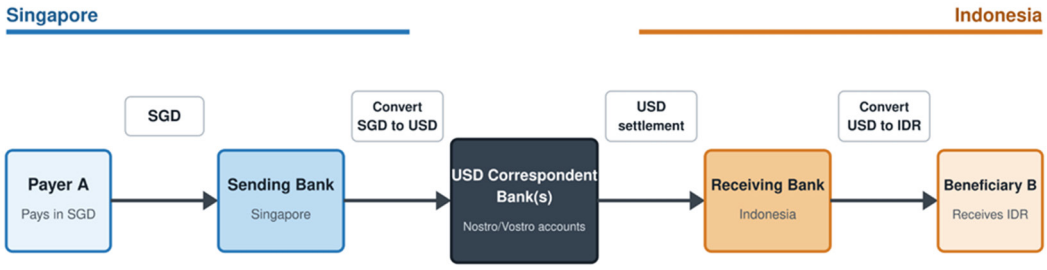
Under the correspondent banking model, a bank maintains an account with another bank in a foreign jurisdiction or currency. The account is referred to as a "Nostro account" from the perspective of the bank holding funds abroad and as a "Vostro account" from the perspective of the bank maintaining the account. When the sending and receiving banks do not have a direct relationship, one or more intermediary banks may be required to complete the payment chain.

A cross-border payment involves both the transmission of payment instructions and the movement of funds across accounts. SWIFT provides standardized and secure financial messaging, but it does not itself hold funds or settle transactions. Settlement takes place through debits and credits to correspondent accounts and, where relevant, through domestic central bank or commercial bank settlement systems. Because messaging and settlement remain institutionally separate, each intermediary may need to validate payment information, conduct compliance checks, and reconcile the transaction before passing it to the next institution.

Correspondent banking provides broad international reach, access to foreign-currency liquidity, established legal arrangements, and mechanisms for compliance and risk management. Its multilayered structure nevertheless creates significant frictions. Payments may be delayed by differences in operating hours and time zones, incomplete or inconsistent data, manual intervention, repeated AML/CFT checks, sanctions screening, and the involvement of multiple intermediaries. Fees, foreign exchange costs, and the status of a payment may also be difficult for the originator and beneficiary to identify in advance. Maintaining liquidity in foreign-currency Nostro accounts also ties up balance-sheet resources and may increase funding costs. These problems correspond broadly to the challenges of cost, speed, access, and transparency identified under the G20 cross-border payments agenda (FSB 2021).

Correspondent banking relationships have also become more concentrated as some global banks have withdrawn from jurisdictions or customer segments considered costly or difficult to serve. Such de-risking may reduce access to cross-border payment services, particularly for smaller financial institutions and less-connected economies.

Figure 7: Conventional Cross-border Payment through Correspondent Banking



Note: Illustrative example of a conventional payment from Singapore to Indonesia through the US dollar.
Source: Author's illustration based on BIS (2026a).

Correspondent banking should not, however, be treated simply as an outdated technology to be replaced. It combines payment execution with liquidity provision, foreign exchange services, regulatory compliance, and legally recognized settlement arrangements. The policy challenge is therefore to improve its speed, transparency, and efficiency without weakening financial integrity, settlement finality, or operational resilience.

Figure 7 illustrates a conventional payment from Singapore to Indonesia in which the foreign exchange transaction and interbank settlement are intermediated through the US dollar. When the sending and receiving banks do not have a direct correspondent relationship, one or more intermediary banks may transmit payment instructions, adjust correspondent-account balances, conduct compliance checks, and deduct applicable fees before the beneficiary's bank receives the funds.

Local Currency Transaction and Settlement Frameworks

Against this background, a growing number of Asian economies have introduced local currency settlement (LCS) or broader local currency transaction (LCT) frameworks. Earlier LCS arrangements focused principally on settling bilateral trade and investment in the currencies of the participating economies. More recent LCT frameworks also cover foreign exchange transactions, local-currency liquidity, hedging instruments, and a wider range of eligible transactions. In this paper, LCT/LCS frameworks refer collectively to arrangements that facilitate bilateral transactions in participating currencies rather than systematically relying on a third-country vehicle currency.

These frameworks aim to reduce foreign exchange conversion costs, limit exposure to movements in the US dollar, deepen regional currency markets, and strengthen payment resilience. They do not necessarily eliminate the use of the US dollar or establish a new payment infrastructure. Instead, participating authorities designate commercial banks—known in Indonesia's arrangements as appointed cross-currency dealers—to quote the two local currencies directly, maintain accounts in the partner currency, and provide foreign exchange, liquidity, and hedging services (Bank Indonesia 2017).

A Singapore–Indonesia transaction, for example, may be converted directly between Singapore dollars and Indonesian rupiah, rather than through successive Singapore dollar–US dollar and US dollar–rupiah conversions. Payment execution may nevertheless continue through correspondent accounts and relevant domestic payment systems. LCT/LCS frameworks therefore primarily modify the currency-conversion arrangements within bank-based cross-border payments. Their effectiveness depends on competitive quotations, sufficient market liquidity, access to hedging instruments, and firms' willingness to invoice and hold balances in the participating currencies

The development of LCT/LCS arrangements in Asia has proceeded mainly through bilateral agreements rather than through a single region-wide settlement platform. Indonesia, Malaysia, and Thailand formed the initial core of these arrangements. In February 2025, their central banks introduced harmonized operational guidelines that consolidated their existing bilateral frameworks, streamlined operational procedures, and added portfolio investment to the eligible transactions already covering trade and direct investment (BOT, Bank Indonesia, and BNM 2025).

Indonesia's bilateral LCT partnerships have subsequently expanded to include Japan, the People's Republic of

Table 1: Major Bilateral Local Currency Transaction/Local Currency Settlement Arrangements in Asia

Economies	Main Features
Indonesia–Malaysia	Early ASEAN arrangement using rupiah and ringgit for trade, investment, and subsequently a wider range of transactions.
Indonesia–Thailand	Direct rupiah–baht transactions through designated banks; part of the harmonized ASEAN LCT framework.
Malaysia–Thailand	Facilitates bilateral transactions in ringgit and baht and forms part of the core ASEAN local-currency network.
Indonesia–Japan	Rupiah–yen settlement, expanded from trade and direct investment to a broader range of bilateral transactions.
Indonesia–PRC	Supports direct rupiah–renminbi transactions and has become one of Indonesia's largest LCT corridors.
Indonesia–Republic of Korea	Designated banks provide direct rupiah–won quotations and settlement services.
Indonesia–Singapore	Operationalized on 31 August 2026 for bilateral trade transactions using rupiah and Singapore dollars.

ASEAN = Association of Southeast Asian Nations, Lao PDR = Lao People's Democratic Republic, LCS = local currency settlement, LCT = local currency transaction, PRC = People's Republic of China.

Sources: Bank Indonesia (2017, 2025a), BOT, Bank Indonesia, and BNM (2025), Bank Indonesia and MAS (2026), and Greater Mekong Subregion Secretariat (2025).

China, the Republic of Korea, the United Arab Emirates, and Singapore. Local-currency settlement initiatives have also developed elsewhere in the region, including between Viet Nam and the Lao People's Democratic Republic. The Indonesia–Singapore framework, operationalized on 31 August 2026, is one of the most recent additions and initially focuses on bilateral trade transactions using rupiah and Singapore dollars (Bank Indonesia and MAS 2026). Table 1 presents selected operational bilateral frameworks and their principal features; it is not intended to provide an exhaustive inventory of all local-currency arrangements in Asia.

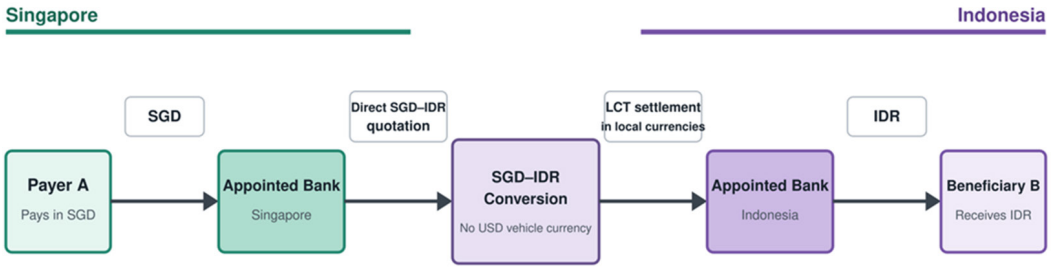
Indonesia has become the most active hub of bilateral LCT arrangements in Asia. The individual frameworks nevertheless differ in their starting dates, eligible underlying transactions, participating financial institutions, documentation requirements, and operational rules. The harmonized framework among Indonesia, Malaysia, and Thailand has introduced greater consistency across these three economies, but substantial differences remain across the other bilateral corridors.

Available evidence indicates that the use of LCT arrangements is increasing, although comparable data for individual currency corridors remain limited. Indonesia provides the clearest aggregate evidence. The value of its LCT transactions with partner economies rose from approximately $4.7 billion in the first half of 2024 to $11.7 billion in the first half of 2025, an increase of about 149% (Bank Indonesia 2025a, 2025b). These figures should be interpreted carefully because the increase reflects not only greater use of established corridors but also the addition of partner economies and the expansion of eligible underlying transactions. It therefore does not represent the growth of a constant set of bilateral arrangements.

The Indonesia–Singapore framework is too recent to provide a meaningful time series. It nevertheless offers a useful example of how an LCT framework changes the foreign exchange component of a conventional correspondent-banking payment. As illustrated in Figure 8, a payer in Singapore instructs its bank to make a trade payment to a beneficiary in Indonesia. Participating banks designated under the framework can quote Singapore dollars against rupiah directly and process the payment through accounts maintained with their designated counterpart banks. The payer can provide Singapore dollars while the beneficiary receives rupiah, without requiring the payment amount to be converted first into US dollars and subsequently into rupiah (Bank Indonesia and MAS 2026).

Compared with the conventional transaction illustrated in Figure 7, the principal innovation in Figure 8 is the shortening of the foreign exchange chain rather than the elimination of correspondent banking. Direct Singapore dollar–rupiah conversion may reduce conversion spreads and exposure to movements in the US dollar, but the benefits depend on sufficient liquidity, competitive quotations, access to hedging instruments, and transaction volumes. The Indonesia–Singapore arrangement is therefore best understood as an

Figure 8: Singapore–Indonesia Local Currency Transaction Framework with Direct Singapore Dollar–Rupiah Conversion



LCT = local currency transaction.

Source: Author's illustration based on Bank Indonesia and Monetary Authority of Singapore (2026).

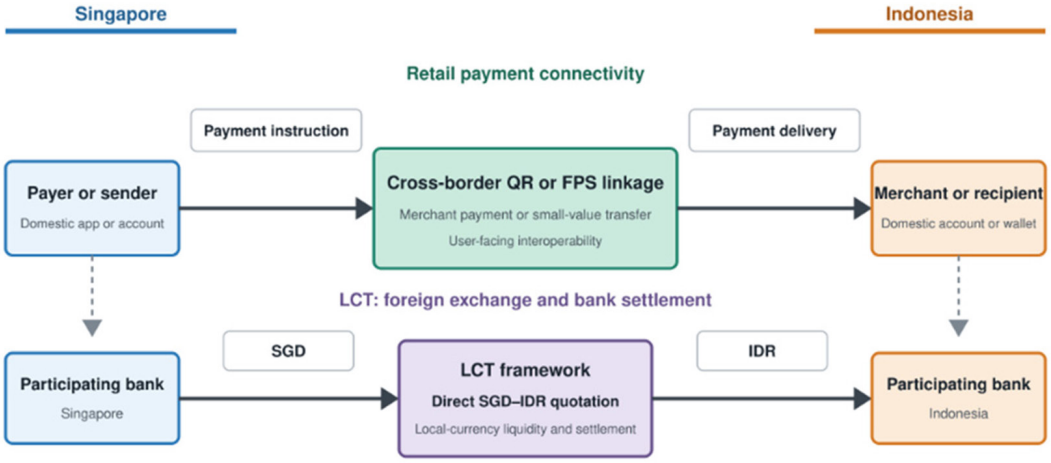
incremental modernization of bank-based cross-border payments rather than a new settlement infrastructure.

Relationship between LCT Frameworks and Cross-border Retail Payment Connectivity

LCT frameworks are related to, but distinct from, the expansion of cross-border retail payment connectivity in Asia. Cross-border QR and fast payment system linkages provide the user-facing infrastructure through which consumers, merchants, and individuals initiate merchant payments or small-value transfers. LCT frameworks operate at a different layer by providing participating banks with arrangements for direct local-currency quotation, foreign exchange conversion, liquidity provision, and interbank settlement.

The two layers can nevertheless be complementary. Retail payment linkages can improve the accessibility and speed of cross-border payments, while LCT frameworks may reduce reliance on a third-country vehicle currency in the underlying foreign exchange and settlement process. An LCT framework does not by itself create QR interoperability or person-to-person transfer functionality, and a retail payment linkage does not necessarily rely on a formal LCT arrangement. Figure 9 illustrates the distinction and potential interaction between these two layers.

Figure 9: Relationship between Retail Payment Connectivity and Local Currency Transaction Frameworks



LCT = local currency transaction.

Source: Author's illustration based on ASEAN (2023) and Bank Indonesia and MAS (2026).

3.2 Modernizing Correspondent Banking through SWIFT GPI and ISO 20022

Wholesale payment reform is developing along a continuum rather than through the simple replacement of correspondent banking. At one end, banks and payment infrastructure are improving the speed, transparency, and data quality of existing cross-border payment chains. At the other, central banks and regulated financial institutions are testing tokenized forms of commercial bank money, central bank money, and financial assets on shared digital platforms. These approaches address related but different frictions and are likely to coexist for a considerable period.

SWIFT GPI improves the transparency and management of correspondent-banking payments by assigning each transaction a unique end-to-end reference and enabling participating banks—and corporate customers through their banks—to track a payment from initiation to confirmation. Better information about payment status, routing, fees, and the time at which funds are credited to the beneficiary can reduce uncertainty, accelerate investigations, and facilitate invoice reconciliation. However, GPI operates through existing correspondent-banking arrangements. It does not itself remove intermediary banks, Nostro and Vostro accounts, foreign exchange transactions, or the liquidity required to complete settlement (SWIFT n.d.b).

The adoption of ISO 20022 complements these improvements by providing richer, more structured, and machine-readable payment data. Standardized information on the parties, transaction purpose, remittance details, and other payment elements can support straight-through processing, sanctions screening, AML/CFT controls, fraud monitoring, and automated reconciliation. For trade-related payments, richer data may also make it easier to link a payment with an invoice, purchase order, shipment, or other commercial record. These benefits depend on consistent implementation and end-to-end data quality. If intermediaries truncate information, domestic systems apply different data requirements, or firms cannot map their internal data to the standard, a common message format alone will not achieve full interoperability (SWIFT n.d.a).

SWIFT GPI and ISO 20022 can therefore reduce information, tracking, and processing frictions while leaving the underlying economic structure of correspondent banking largely intact. Banks must still maintain accounts and liquidity in the relevant currencies, manage intraday funding, conduct customer due diligence and sanctions screening, and comply with different legal and regulatory requirements. Smaller institutions and less frequently used currency corridors may also continue to face high costs because they depend on longer correspondent chains and relatively shallow foreign exchange markets. These limitations help explain the growing interest in tokenized money and shared settlement platforms examined in the next subsection.

3.3 Tokenized Wholesale Settlement: Project Agorá and Wholesale Central Bank Digital Currency

Tokenization and wholesale central bank digital currency represent closely related but institutionally distinct approaches to the modernization of wholesale payment infrastructure. Both seek to place regulated forms of money and settlement assets on programmable digital platforms, potentially allowing payment instructions, foreign exchange transactions, compliance processes, and final settlement to be coordinated more directly. Their institutional designs nevertheless differ. Project Agorá combines tokenized commercial bank deposits with tokenized central bank reserves within the existing two-tier monetary system, whereas wholesale CBDC and multi-CBDC arrangements focus more directly on the issuance, use, and interoperability of digital central bank liabilities among participating financial institutions. This subsection first explains the common technological and monetary foundations of these initiatives, then examines Project Agorá and wholesale CBDC experiments, before considering their operational, legal, governance, and interoperability challenges.

Tokenized Money and the Two-tier Monetary System

Tokenization refers to the generation and recording of a digital representation of an asset or financial claim on a programmable platform. In wholesale finance, tokens may represent commercial bank deposits, central bank reserves, securities, or other regulated claims. They should be distinguished from unbacked cryptocurrencies. A tokenized deposit remains a liability of the issuing commercial bank, and a tokenized security remains a claim governed by the applicable legal and regulatory framework. Tokenization changes how a claim is recorded and transferred; it does not automatically change the identity or obligations of its issuer (FSB 2024).

This distinction is particularly important in wholesale payments because the existing monetary system is based on a two-tier structure. Commercial banks provide deposit money and payment services to their customers, while central banks provide the ultimate settlement asset used between financial institutions. Tokenization can preserve this institutional division while changing the technological infrastructure through which commercial bank money and central bank money are issued, transferred, and settled.

Under such an arrangement, commercial banks may issue tokenized deposits that remain claims on those banks, while central banks may issue tokenized reserves or wholesale central bank money that provides the settlement asset for participating financial institutions. Bringing both forms of regulated money onto coordinated platforms could preserve the customer-facing and credit-intermediation roles of commercial banks while retaining the safety and finality associated with central bank money. Tokenized wholesale settlement should therefore not necessarily be interpreted as an attempt to replace commercial bank money with either privately issued cryptocurrency or universally accessible central bank money.

Programmability and Atomic Settlement

A shared programmable platform can coordinate the record of monetary claims, payment instructions, and the rules governing their transfer. Financial institutions would no longer need to reconcile every stage of a transaction across a sequence of separate bilateral ledgers in the same way as under conventional correspondent banking. Programmable functions can embed workflow logic, compliance requirements, and conditional payment triggers into the transaction process.

The objective of a programmable platform is not necessarily to replace existing financial institutions. Rather, it is to reduce the sequential processing, manual intervention, and uncertainty that arise when payment instructions, foreign exchange conversion, compliance checks, and final settlement are handled through separate systems and institutions. Where the required information and settlement assets are available on interoperable platforms, several stages of a transaction could be coordinated more closely.

A central technical feature of these arrangements is "atomic settlement." Under atomic settlement, linked components of a transaction are completed together or not completed at all. In a foreign exchange transaction, payment-versus-payment can ensure that one currency is transferred only if the counter-currency is also transferred, thereby reducing principal risk. Similar coordination can support delivery-versus-payment arrangements in which the transfer of a security or another financial asset occurs only if the corresponding payment is completed.

Atomicity may also reduce the period during which liquidity or collateral remains tied up. Its net effect on liquidity is not automatically positive, however. The outcome depends on transaction sequencing, access to intraday funding, the timing of settlement, and whether participants can offset or net their payment obligations. A system that settles every transaction individually and immediately may require participants to hold more intraday liquidity unless appropriate liquidity-saving mechanisms are incorporated.

Programmability could also connect payment with the conditions of an underlying commercial transaction. A payment might be released when specified documentation has been verified, when an asset has been delivered, or when contractual conditions recorded on the platform have been met. For trade-related transactions, this creates a potential link among tokenized money, electronic invoices, trade documents, logistics data, and supply-chain finance.

The principal economic gain would come from reducing the repeated verification and reconciliation that currently separate the commercial transaction, compliance review, financing, foreign exchange conversion, and final settlement. These broader trade applications nevertheless require reliable external data, legally recognized electronic documents, and effective mechanisms for correcting inaccurate or disputed information. They are not produced by tokenizing money alone.

Project Agorá and Hybrid Tokenized Money

Project Agorá, convened by the Bank for International Settlements (BIS) Innovation Hub in collaboration with the Institute of International Finance, examines how tokenized commercial bank deposits and tokenized central bank money could be integrated on a shared programmable platform for wholesale cross-border payments. The project brought together eight central banks and more than 40 regulated financial institutions and developed a prototype rather than a live payment system (BIS Innovation Hub 2026).

The project is designed to preserve the two-tier monetary system. Commercial bank deposits would remain liabilities of individual banks, while tokenized central bank money would provide the ultimate settlement asset. By bringing both forms of money onto a shared platform, Project Agorá explores whether payment instructions, foreign exchange transactions, compliance checks, and final settlement can be coordinated more closely while retaining the respective roles of commercial banks and central banks.

The prototype examined cross-border transfers of tokenized deposits and atomic multi-currency settlement, under which all components of a transaction are completed simultaneously or none occurs. This could reduce principal risk and allow compliance checks, conditional payment instructions, and other transaction workflows to be incorporated into the settlement process.

At scale, such a platform could reduce sequential processing and support longer operating hours and multi-currency transactions. However, atomic settlement does not eliminate the need for adequate liquidity, and participating institutions would still need to obtain and manage funds in the relevant currencies.

Project Agorá remains experimental. Implementation would require legal clarity regarding tokenized money and settlement finality, agreement on access and governance, effective compliance and data-protection arrangements, operational resilience, sufficient liquidity, and interoperability with existing bank accounts, real-time gross settlement systems, correspondent banking networks, and domestic legal frameworks. Its significance therefore lies in testing whether tokenization and programmability can modernize the two-tier banking system rather than replace it.

Wholesale CBDC and Multi-CBDC Arrangements

A related area of experimentation concerns wholesale central bank digital currency. Unlike retail CBDC, which is intended to be accessible to households and businesses, wholesale CBDC is generally designed for use by authorized financial institutions conducting interbank, securities, or foreign exchange transactions. It allows existing sovereign currencies to be used as settlement assets in digital and potentially programmable environments.

An important potential application is cross-border payment-versus-payment settlement. If the wholesale CBDCs of two jurisdictions can be transferred on the same platform or across interoperable platforms, the two currency legs of a foreign exchange transaction could be settled simultaneously. This could reduce principal risk, shorten settlement times, and reduce reliance on long chains of correspondent banks and pre-positioned balances. Such arrangements would not, however, eliminate the need for liquidity in the participating currencies.

One prominent example is mBridge, a multi-CBDC platform originally developed through cooperation involving the BIS Innovation Hub and the central banks or monetary authorities of the PRC; Hong Kong, China; Thailand; and the United Arab Emirates. The platform was tested through a pilot involving real-value transactions in 2022 and reached the minimum viable product stage in 2024. The Saudi Central Bank became a full participant in 2024 but concluded its participation after completing its proof-of-concept work in May 2025. The BIS Innovation Hub also concluded its direct involvement in 2024 and handed the project over to the participating central banks, which have continued its development (BIS Innovation Hub 2022, 2024a; CBUAE 2026; Financial Times 2026).

mBridge differs institutionally from Project Agorá. Project Agorá combines tokenized commercial bank deposits with tokenized central bank money, preserving the respective roles of commercial banks and central banks within the two-tier monetary system. By contrast, mBridge places greater emphasis on the cross-border transfer of wholesale CBDCs issued by participating central banks. Commercial banks continue to perform customer due diligence, payment initiation, foreign exchange, and compliance functions, but settlement on the platform takes place in central bank money rather than through a chain of correspondent-bank claims.

Bilateral CBDC arrangements represent another possible model. In 2023, the Reserve Bank of India and the Central Bank of the United Arab Emirates agreed to explore interoperability between their CBDCs and to conduct proof-of-concept and pilot work on a bilateral CBDC bridge for cross-border remittances and trade. Bilateral initiatives may be less institutionally complex than a large multi-CBDC platform and may allow jurisdictions with substantial economic links to test currency conversion, settlement finality, and regulatory coordination in a more limited environment (CBUAE and RBI 2023).

These initiatives could diversify regional settlement options, but they do not imply the imminent replacement of the dollar-based international financial system. The US dollar, dollar liquidity, correspondent banking networks, and dollar-denominated financial markets remain deeply embedded in global trade and finance. Transactions between regional currencies may also continue to be priced, funded, or hedged through the US dollar even if final settlement can take place directly in the participating currencies. The more likely near-term outcome is therefore the coexistence of correspondent banking, bilateral local-currency arrangements, and tokenized or multi-CBDC settlement platforms.

Legal, Governance, and Interoperability Challenges

Tokenized deposits and wholesale CBDC arrangements face legal, governance, and operational challenges that cannot be resolved through technology alone. Atomic settlement requires the different legs of a transaction to be coordinated on the same platform or across interoperable platforms. Technical completion does not by itself guarantee legal settlement finality. Participating jurisdictions must clarify the legal status of tokenized deposits and wholesale CBDC, the point at which transfers become irrevocable, and the treatment of digital claims if a participating institution fails.

Cross-border platforms also require clear governance arrangements. Authorities must determine which institutions can participate, who controls admission and exclusion, how platform rules can be changed, and how decisions are made during operational or financial emergencies. Responsibilities for AML/CFT controls, sanctions compliance, cybersecurity, data protection, and operational failures must also be allocated among central banks, commercial banks, and platform operators. Differences in data-localization, confidentiality, and public-authority access rules may further complicate cross-border information sharing (FSB 2024).

Interoperability will be critical because tokenized platforms are unlikely to replace existing infrastructure in a single transition. They will need to interact with correspondent banking networks, domestic real-time gross settlement systems, commercial banks' core systems, foreign exchange markets, securities settlement systems, and trade-data platforms. ISO 20022 can support this interaction by providing structured and standardized payment information, but common messaging standards cannot by themselves resolve differences in legal frameworks, participation rules, settlement assets, liquidity arrangements, and operating hours.

Tokenized settlement arrangements should therefore be assessed not only by their technical performance but also by whether they can establish legal certainty, trusted governance, adequate liquidity, operational resilience, and interoperability across jurisdictions. If different groups of countries or institutions develop incompatible platforms, financial institutions may have to maintain liquidity, technology, and compliance processes across several networks. Tokenization could then reproduce the fragmentation of correspondent banking in a new technological form rather than overcome it.

3.4 Toward a Multilayered Wholesale Settlement Architecture

Wholesale payment infrastructure is evolving through several parallel approaches rather than a single technological transformation. SWIFT GPI and ISO 20022 are improving the speed, transparency, and data quality of established correspondent banking. Local currency transaction frameworks allow selected bilateral trade and financial transactions to be invoiced and settled in participating currencies, although they may continue to use correspondent banks and SWIFT messaging. Tokenized deposits and wholesale CBDC platforms, meanwhile, are exploring new ways of coordinating payment, foreign exchange, compliance, and final settlement.

These arrangements are more likely to coexist than to replace one another. Correspondent banking provides broad global reach and established legal and institutional relationships. Local currency arrangements can reduce reliance on an intermediary currency in selected corridors. Tokenized deposit platforms may improve coordination while preserving the two-tier banking system, while wholesale CBDC and multi-CBDC platforms may provide additional settlement options based directly on central bank money. For Asian economies, which differ substantially in financial development, currencies, capital-account regimes, regulatory capacity, and payment infrastructure, no single model is likely to meet the needs of every country or transaction.

The central policy challenge is therefore to ensure that these different arrangements can interact without creating additional liquidity silos, legal uncertainty, or operational fragmentation. This requires legal settlement finality, adequate liquidity, effective governance, cyber

and operational resilience, coordinated compliance responsibilities, and interoperability between existing and emerging systems. The value of wholesale modernization will ultimately depend not only on technical performance but also on whether it can support trade and other economic activity while remaining connected to established domestic and international financial infrastructure.

4. Trade Payments, Digitalization, and Small and Medium-sized Enterprise Participation

Sections 2 and 3 examined the retail and wholesale layers of cross-border payment infrastructure. This section shifts the focus from payment systems themselves to their use in international trade, examining how payments interact with commercial contracts, electronic trade documents, logistics and customs data, foreign exchange services, and financing, with particular attention to the constraints faced by SMEs. Its central argument is that faster payment alone does not necessarily improve SME participation in international trade: the economic benefits depend on whether payment innovation can be connected to legally recognized electronic trade documents, reliable transaction data, affordable foreign exchange services, and access to working-capital and trade-finance facilities. Retail payment connectivity can facilitate small-value cross-border sales and payment collection, while wholesale modernization can improve the banking and settlement infrastructure supporting larger business transactions. However, neither layer by itself resolves the financing, documentation, compliance, and data-related constraints that frequently prevent SMEs from trading across borders.

4.1 From Commercial Transaction to Final Settlement

A cross-border trade transaction involves more than the transfer of funds between a buyer and a seller. It generally begins with an order and commercial contract, followed by production, invoicing, shipment, customs clearance, delivery, payment, and financial reconciliation. Depending on the transaction, it may also involve foreign exchange conversion, insurance, trade credit, guarantees, and financing provided before or after shipment.

Financial institutions support these activities through several types of trade-finance arrangements. Under a letter of credit, a bank commits to pay an exporter if the exporter presents the documents specified in the contract. Documentary collection typically provides a less costly alternative, under which banks transmit trade documents and collection instructions without guaranteeing payment. In open-account trade, goods are shipped before payment is received. Exporters may therefore rely on receivables finance, factoring, supply-chain finance, or other forms of working-capital support.

These arrangements address a basic timing mismatch in international trade. Exporters must finance materials, labor, production, transportation, and insurance before receiving payment, while importers may prefer to pay only after the goods have been shipped, inspected, or delivered. This gap can be especially difficult for SMEs, which must finance operating costs between production and the receipt of funds. A faster cross-border payment system can shorten the final payment stage, but it does not automatically provide the pre-shipment or post-shipment finance required to complete the underlying trade.

Trade payments also depend on the infrastructure examined in Sections 2 and 3. A small online transaction may be initiated through a fast payment system, an account-based transfer, or a digital wallet, while a cross-border QR arrangement may be used for an in-person merchant payment. A larger B2B payment will generally involve commercial banks, correspondent banking relationships, and wholesale settlement systems, as well as foreign exchange services when currency conversion is required. Even when the commercial order and invoice are generated digitally, the financial component may continue to rely on conventional bank accounts, compliance procedures, and settlement infrastructure.

The distinction between retail and wholesale payments should therefore be understood in relation to the economic function and structure of the transaction rather than the identity of the firm alone. SMEs may use retail payment connectivity for small-value e-commerce sales, tourism-related services, or payments received through digital platforms. The same firms may rely on commercial banks and wholesale payment infrastructure when paying foreign suppliers, converting currencies, or settling larger B2B transactions, while also requiring working-capital or trade-finance facilities to finance production and inventories.

4.2 SME Constraints in Cross-border Trade and Finance

SMEs face disproportionate frictions in cross-border trade because many costs associated with payments, compliance, documentation, and credit assessment are fixed at the transaction or customer level. Verifying a new customer, screening a payment, examining trade documents, and conducting due diligence can require substantial resources regardless of the size of the transaction. These costs therefore impose a heavier relative burden on smaller firms and transactions.

SMEs are also more likely to have limited collateral, short credit histories, incomplete financial statements, and less established relationships with financial institutions. Banks may consequently find it difficult to assess the creditworthiness of a small exporter or verify the underlying commercial transaction. These information problems become more severe when buyers, sellers, and financial institutions are located in different jurisdictions and operate under different legal, accounting, and regulatory systems.

The resulting trade–finance gap remains substantial. The Asian Development Bank (ADB) estimated that global unmet demand for trade finance remained at approximately $2.5 trillion in 2025, equivalent to about 10% of global trade (ADB 2025). This estimate is not exclusively an SME financing gap, but smaller firms are particularly affected because they often have fewer financing alternatives and may find available products too costly, collateral-intensive, or administratively burdensome. Faster payment cannot resolve this problem by itself: an SME may be able to receive funds more quickly after a sale while remaining unable to finance production and shipment before payment is received.

Payment and foreign exchange costs create additional constraints. SMEs generally have less bargaining power than multinational corporations when negotiating bank fees, foreign exchange spreads, and payment terms. They may also have limited access to competitive foreign exchange quotations and affordable hedging instruments. In less liquid currency corridors, payments may be converted through the US dollar or another major currency, adding an additional foreign exchange transaction. Local currency transaction arrangements may reduce some of these costs by facilitating direct quotations between participating currencies, but their benefits depend on market liquidity, competition among providers, access to hedging instruments, and the transmission of lower bank-level costs to SME customers.

Smaller firms may also depend on domestic banks with limited international networks. Their payments may therefore pass through several correspondent institutions, increasing fees, processing time, and the possibility of additional compliance reviews. AML/CFT requirements, sanctions screening, customer due diligence, and transaction monitoring are necessary to protect the integrity of the financial system, but their fixed costs can discourage financial institutions from serving unfamiliar customers, smaller firms, or less profitable currency corridors. Proportionate and risk-based compliance is therefore important to avoid excluding legitimate firms from cross-border financial services.

Lower-cost digital payments can address part of these constraints. Cross-border fast payment linkages and account-based transfers can facilitate small-value transfers and payment collection, while QR connectivity can support in-person payments to SME merchants. However, payment connectivity alone does not provide working capital, establish the validity of trade documents, or give lenders sufficient information to assess credit risk. Its wider contribution to SME trade will depend on whether payment information can be connected securely with commercial, logistics, customs, and financing data.

Experience in Asia illustrates both the potential and the current limits of this connection. ADB's Trade and Supply Chain Finance Program supported approximately 15,000 SMEs in cross-border transactions in 2025 through guarantees and loans, demonstrating that access to trade finance can be expanded through financial institutions (ADB 2026). The International Finance Corporation (IFC) has also supported a cross-border supply-chain finance initiative intended to give SME suppliers in several Asian economies earlier access to payment on their invoices (IFC 2024). These initiatives address financing and payment timing, but they do not establish that cross-border QR or fast payment linkages have themselves increased access to trade finance. Connecting small-value payment records with verified orders, invoices, and shipment data remains a promising avenue for future development, subject to appropriate data-sharing and credit-assessment arrangements.

4.3 Connecting Payment Data with Trade and Logistics Data

The economic value of digitalization may arise not only from faster payment but also from connecting payment information with commercial, logistics, customs, and

financing data. A cross-border transaction can generate an order, contract, invoice, transport document, insurance record, customs declaration, delivery confirmation, payment instruction, and settlement record. These records are often maintained in separate systems operated by firms, banks, logistics companies, insurers, customs authorities, and digital platforms.

This fragmentation leads to repeated data entry, document verification, and reconciliation. A bank may receive an invoice without being able to confirm whether the goods were shipped, while a logistics provider may hold delivery information that cannot readily be matched with the relevant invoice or payment. Differences in data formats and identifiers also make it more difficult to verify genuine transactions and detect fraud, duplicate invoices, or multiple financing of the same receivable.

Greater interoperability could allow verified commercial activity to generate a more usable digital transaction record. With appropriate authorization, a lender might confirm that an order was placed, goods were shipped, customs procedures were completed, and previous payments were made as agreed. Such information could complement conventional financial statements and collateral when assessing credit risk. SMEs with limited credit histories, reliable records of orders, invoices, deliveries, and payments could form part of a broader transaction-based credit profile and potentially support access to receivables or supply-chain finance.

The legal recognition of electronic trade documents is essential to this process. Some documents perform more than an information function. A bill of lading, for example, may represent rights relating to the goods and must be transferable between parties. If applicable law recognizes only a paper original, digital data exchange cannot fully replace the underlying paper-based process. The United Nations Commission on International Trade Law's Model Law on Electronic Transferable Records provides a framework under which electronic transferable records can be treated as functionally equivalent to paper documents when requirements concerning integrity, control, and reliability are satisfied (UNCITRAL 2017). Wider adoption and cross-border recognition of such frameworks could support paperless trade and closer integration among documentation, payment, and finance.

Programmable payments could further strengthen this connection by allowing payment to be released when specified conditions are satisfied—for example, when an authorized electronic document is presented or trusted data confirm that goods have reached an agreed stage of the transaction. Where the legal and technical arrangements permit, payment and the transfer of a digital trade document or financial asset could be coordinated more closely. Such arrangements should nevertheless be treated cautiously. Automated execution is only as reliable as the underlying data and contractual conditions. Incorrect logistics data, fraudulent invoices, disputed deliveries, or malfunctioning external data sources could cause payment to be released or withheld improperly.

Data integration also creates risks involving privacy, cybersecurity, market power, and financial exclusion. Firms need to know how their transaction data are collected, shared, and used, while cross-border data restrictions may limit access to relevant information. SMEs may become dependent on proprietary platforms and may be unable to transfer their commercial histories to another provider. Data-based credit models may also disadvantage firms that have limited digital records, operate partly in cash, or are located in areas with weak digital infrastructure. Policies should therefore promote common data and identity standards, data portability, informed consent, accountability, and proportionate access. Connecting payment and trade data can support SME finance, but only if reliable digital records are legally recognized and can be shared securely across institutions and jurisdictions.

4.4 Complementary Roles of Retail and Wholesale Payment Innovation in Trade

Retail and wholesale payment reforms affect international trade through different but complementary channels. Retail connectivity improves the interface through which consumers, individuals, merchants, and smaller firms initiate and receive cross-border payments. Wholesale modernization affects the banking relationships, foreign exchange arrangements, and settlement systems that support larger payments and interbank obligations.

Retail connectivity is especially relevant to small-value commerce. Cross-border fast payment linkages and account-based transfers can support payment collection, small-value transfers, and remittances, while QR connectivity can facilitate in-person payments to merchants. By allowing users to make payments through familiar domestic applications and bank accounts, these arrangements can reduce payment friction and expand the potential customer base of smaller firms. However, retail connectivity does not eliminate the need for foreign exchange conversion, compliance, and settlement

between participating financial institutions. Merchant QR payments also do not provide the invoicing, working-capital finance, or risk-mitigation functions required for many B2B transactions.

Wholesale modernization plays a different role. Improvements in payment messaging, transaction tracking, and ISO 20022 can reduce uncertainty and facilitate reconciliation by providing richer and more structured information. Local currency transaction arrangements may reduce reliance on a third-country currency in selected corridors. Tokenized deposits and wholesale CBDC platforms may eventually allow payment, foreign exchange, compliance, and settlement to be coordinated more closely. These developments can improve the infrastructure supporting trade, but their benefits for SMEs are mainly indirect.

Infrastructure-level efficiency gains do not automatically result in lower charges, better foreign exchange rates, or greater access to trade finance. Banks may settle transactions more quickly while maintaining the same customer fees and credit standards. New tokenized platforms may also be accessible mainly to large international banks, leaving smaller financial institutions and their SME customers outside the network. The distribution of benefits will therefore depend on competition, participation rules, access by smaller institutions, and the extent to which banks pass efficiency gains on to corporate customers.

Policy evaluation should consequently distinguish between technical performance and economic outcomes. Settlement speed, processing capacity, and operational resilience indicate whether an infrastructure functions effectively, but they do not show who ultimately benefits. Evaluation should also consider whether SMEs experience lower total payment costs, narrower foreign exchange spreads, faster access to funds, fewer rejected payments, improved financing terms, and broader access to cross-border services. The contribution of payment innovation to trade will be greatest when retail accessibility and wholesale efficiency are connected to electronic trade documents, digital identity, foreign exchange services, and working-capital and trade-finance facilities.

4.5 Implications for Inclusive Digital Trade

Payment innovation can support more inclusive trade only when it is integrated with the legal, data, and financial infrastructure of cross-border commerce. Faster settlement should therefore be accompanied by the legal recognition of electronic trade documents, interoperable data and identity standards, proportionate compliance, transparent payment and foreign exchange costs, and wider access to working-capital and trade-finance facilities. These priorities form the basis of the sequenced policy agenda presented in Section 5.

5. Policy Priorities and a Sequenced Agenda for Asia

Cross-border payment reform should begin with strong domestic foundations rather than the introduction of a particular technology. Priorities include reliable domestic payment systems, common data and identity standards, legal recognition of electronic trade documents, proportionate compliance, transparent payment and foreign exchange costs, and broad access for regulated financial institutions and payment providers.

The next step should be to expand proven cross-border arrangements where there is clear economic demand. QR merchant payments, fast payment system linkages, account-to-account transfers, improvements to correspondent banking, and Local currency transaction arrangements can be developed progressively along important trade, tourism, and remittance corridors. Regional cooperation should support interoperability without requiring every economy to adopt a single payment model.

Payment connectivity should then be linked with electronic invoices and trade documents, customs and logistics data, foreign exchange services, and working-capital and trade-finance facilities. Tokenized deposits and wholesale CBDC platforms should be pursued selectively where they provide a clear advantage over existing arrangements and where legal certainty, governance, liquidity, operational resilience, and interoperability can be established.

Finally, progress should be evaluated through outcomes for users rather than technical performance alone. Relevant indicators include total transaction costs, foreign exchange spreads, predictability and speed of access to funds, rejected or delayed payments, access to trade finance, and participation by SMEs and smaller financial institutions. The objective is not to replace existing systems with a single new infrastructure, but to create a more interoperable, resilient, and inclusive cross-border payment architecture.

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